

Pooyan Jamshidi

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University of South Carolina (USC)
Computer Science and Engineering Department
Columbia, South Carolina 29208
✉ pjamshid@cse.sc.edu
<https://pooyanjamshidi.github.io>

RESEARCH INTERESTS

My research interests span the areas of **Software**, **Systems**, **AI/ML**, and **Robotics**. In particular, I am interested in developing algorithms and tools that enable building resilient systems, deployed in dynamic environments that can automatically handle goal tradeoffs, incorporate user preferences and constraints, identify causes of failures, and self-adapt to recover from failure. I integrate *distributed systems, control theory, statistical learning and optimization, causality, and program analysis*.

PROFESSIONAL EXPERIENCE

- 8/2018–* **Assistant Professor (Computer Science & Engineering)**, *University of South Carolina*, Columbia, SC, US
Directing the AISys lab: <https://pooyanjamshidi.github.io/AISys/>
- 5/2021–12/2021 **Visiting Researcher**, *Google*, Mountain View, CA, US (remote)
Worked with Sugato Basu and Garima Pruthi in the AdsAI team (led by Sugato Basu) on Causal Representation Learning, ML Security, and Self-Supervised Learning.
- 12/2016–8/2018 **Postdoctoral Associate**, *Carnegie Mellon University*, Pittsburgh, US
Worked with Christian Kästner (postdoc advisor) on performance analysis of highly configurable software, collaborated very closely with Norbert Siegmund. Worked with David Garlan on meta-learning for self-adaptive systems. Involved in BRASS MARS, a DARPA-sponsored project, developed transfer learning methods for enabling runtime adaptations of mobile robotics in dynamic environments.
- 2/2015–12/2016 **Postdoctoral Associate**, *Imperial College London*, London, UK
Worked on the theory of uncertainty estimation and Bayesian optimization and applied it to performance auto-tuning of big-data systems. Involved in two EU projects (DICE and MODAClouds), where I developed tools for auto-tuning Apache Hadoop and Storm.
- 9/2014–2/2015 **Postdoc**, *Dublin City University*, Dublin, Ireland
Worked with Claus Pahl on developing self-learning controllers at IC4 cloud center and in collaboration with Intel (Giovani Estrada) and Microsoft (Niall Moran).
- 9/2010–9/2014 **Ph.D. Research Assistant**, *Dublin City University*, Dublin, Ireland
Worked with Claus Pahl (Ph.D. advisor) on developing a cloud controller for auto-scaling in the cloud. I received a scholarship from Lero (the Irish Software Research Centre).

EDUCATION

- 9/2010–9/2014 **Ph.D., Computer Science**, *Dublin City University*, Ireland
○ Thesis: *A framework for robust control of uncertainty in self-adaptive software*
○ Adviser: Prof. Claus Pahl, External examiner: Prof. Pete Sawyer (Lancaster)
- 9/2003–2/2006 **M.Sc., Industrial Engineering (Systems)**, *Amirkabir University of Technology*, Iran
○ Thesis: *An integrated knowledge-based system for product design support*
○ Adviser: Dr. Saeed Mansour
- 9/1999–9/2003 **B.A., Math & Computer Science**, *Amirkabir University of Technology*, Iran

PROFESSIONAL EXPERIENCE: PRE PH.D.

- 7/2008–9/2010 **Research Team Lead (part-time)**, *Shahid Beheshti University*, Tehran, Iran
Led the Automated Software Engineering team in a research lab (Lab Director: Dr. Fereidoon Shams); I have developed and contributed to the development of methods and tools for automating some processes in building distributed systems with service-oriented architectures.
- 2007–2010 **Project Manager**, *Negaran Co*, Tehran, Iran
Managed a large-scale software project on developing a distributed software system automating business processes in a charity organization that has offices in many cities. My team included 20-30 software engineers and designers. Technologies: .NET, Microsoft SQL Server, SOA compatible stacks.
- 2006–2007 **Software Architect**, *Negaran Co*, Tehran, Iran
Developed the enterprise architecture and prepared the ICT business planning using IBM's Business Systems Planning (BSP) and the Zachman Framework. Designed the software architecture of the core systems using an extended ADL in Visual Paradigm.
- 2003–2006 **Software Engineer**, *Rayan Pardaz Kavosh*, Tehran, Iran
Developed server-side software, and designed finance and automation software systems. Added several new features to the code base of a manufacturing automation system. Technologies: Visual C++, COM/CORBA, Socket programming

HONORS AND AWARDS

- UofSC's Breakthrough Stars University of South Carolina's 2022 **Breakthrough Stars Award**. https://sc.edu/uofsc/posts/2022/06/breakthrough_star_pooyan_jamshidi.php
- Junior Researcher Award Computer Science and Engineering Department's **Junior Researcher Award**, University of South Carolina, 2020. <https://cse.sc.edu/news/news/departamental-awards>
- Distinguished Reviewer ACM TOSEM Board of **Distinguished Reviewers**, August 2020. <https://dl.acm.org/journal/tosem/distinguished-reviewers-board>
- Competition Selected as one of the two **finalists** at DCU for a fully funded research visit to IBM Research Brazil, 2014.
- Competition Thesis in 3 **national finalists**, "I bet you didn't know that software can adapt itself on-the-fly", Ireland, 2013, <https://goo.gl/igfKYC>
- Ph.D. Fellowship Awarded Lero Graduate School in Software Engineering (LGSSE) scholarship on the structured Ph.D. in Software Engineering, 2010-2013, **\$85,000**.
- Iranian University Entrance Exam Ranked **21nd** in the national graduate-level exam among 20,000 participants, 2003.

PUBLICATIONS

Key publications are indicated with ★ and top venues are marked with **bold** fonts. 🌐 <http://scholar.google.com/citations?user=41rV5koAAAAJ>

REFEREED JOURNAL ARTICLES

- ★**JSys** [J32] S. Ghafouri, K. Razavi, M. Salmani, A. Sanaee, T. Lorigo-Botran, L. Wang, J. Doyle, P. Jamshidi, *IPA: Inference Pipeline Adaptation to Achieve High Accuracy and Cost-Efficiency*, Journal of Systems Research (JSys), 2023. [revision]
- ★**TOSEM** [J31] V. Cortellessa, J.A. Diaz-Pace, D. Di Pompeo, S. Frank, P. Jamshidi, M. Tucci, A. van Hoorn, *Introducing Interactions in Multi-Objective Optimization of Software Architectures*, ACM Transactions on Software Engineering and Methodology (TOSEM), 2023. [revision]
- ★**JAIR** [J30] M.A. Javidian, O. Pandey, P. Jamshidi, *Scalable Causal Transfer Learning*, Journal of Artificial Intelligence Research (JAIR), 2023. [revision]
- ★**RA-L** [J29] A. Hossen, S. Kharade, B. Schmerl, J. Cámara, J. M. O’Kane, E. C. Czaplinski, K. A. Dzurilla, D. Garlan, P. Jamshidi, *CaRE: Finding Root Causes of Configuration Issues in Highly-Configurable Robots*, IEEE Robotics and Automation Letters (RA-L), 2023 (Also appearing at IROS’23).
- ★**JAIR** [J28] S. Iqbal, J. Su, L. Kotthoff, P. Jamshidi, *FlexiBO: Cost-Aware Multi-Objective Optimization of Deep Neural Networks*, Journal of Artificial Intelligence Research (JAIR), 2023.
- ★**JAIR** [J27] M. Javidian, M. Valtorta, P. Jamshidi, *AMP Chain Graphs: Minimal Separators and Structure Learning Algorithms*, Journal of Artificial Intelligence Research (JAIR), 2020. 🌐 doi:10.1613/jair.1.12101 [SJR rating: **Q1**]
- PCCP** [J26] K. McCullough, T. Williams, K. Mingle, P. Jamshidi, J. Lauterbach, *High-throughput Experimentation meets Artificial Intelligence: A New Pathway to Catalyst Discovery*, Physical Chemistry Chemical Physics (PCCP), 2020. 🌐 doi:10.1039/D0CP00972E [SJR rating: **Q1**]
- ★**ASE** [J25] M. Velez, P. Jamshidi, F. Sattler, N. Siegmund, S. Apel, C. Kaestner, *ConfigCrusher: White-Box Performance Analysis for Configurable Systems*, Springer Automated Software Engineering, 2020. 🌐 doi:10.1007/s10515-020-00273-8 [SJR rating: **Q1**]
- IEEE TSE** [J24] R. Krishna, V. Nair, P. Jamshidi, T. Menzies, *Whence to Learn? Transferring Knowledge in Configurable Systems using BEETLE*, IEEE Transactions on Software Engineering (TSE), 2020. 🌐 doi:10.1109/TSE.2020.2983927 [SJR rating: **Q1**]
- ★**IEEE SOFTWARE** [J23] N. C Mendonça, P. Jamshidi, D. Garlan, and C. Pahl, *Developing Self-Adaptive Microservice Systems: Challenges and Directions*, IEEE Software, 2019. [SJR rating: **Q1**]
- IEEE SOFTWARE** [J22] J. Aldrich, J. Biswas, J. Camáara, D. Garlan, A. Guha, J. Holtz, P. Jamshidi, C. Kaestner, C. Le Goues, A. Mohseni-Kabir, I. Ruchkin, S. Samuel, B. Schmerl, C. Steven Timperley, M. Veloso, and I. Voysey, *Model-based Adaptation for Robotics Software*, IEEE Software, 2019. [SJR rating: **Q1**]
- SPRINGER JBD** [J21] M. Bersani, F. Marconi, D. Tamburri, A. Nodari, P. Jamshidi, *Verifying big data topologies by-design: a semi-automated approach*, Journal of Big Data, 2019. 🌐 doi:10.1186/s40537-019-0199-y [SJR rating: **Q1**]

- IEEE SOFTWARE [J20] C. Trubiani, P. Jamshidi, J. Cito, W. Shang, Z.M. Jiang, M. Borg, *Performance issues? Hey DevOps, mind the uncertainty!*, IEEE Software, 2018. [SJR rating: **Q1**]
- IEEE SOFTWARE [J19] **Featured Article** P. Jamshidi, C. Pahl, N. Mendonça, J. Lewis, S. Tilkov, *Microservices: The Journey So Far and Challenges Ahead*, IEEE Software, 2018. doi:10.1109/MS.2018.2141039, [SJR rating: **Q1**]
- WILEY SPE [J18] A. Balalaie, A. Heydarnoori, P. Jamshidi, D.A. Tamburri, T. Lynn, *Microservices migration patterns*, Wiley Software: Practice and Experience (SPE), 2018. doi:10.1002/spe.2608 [SJR rating: **Q2**]
- ELSEVIER JSS [J17] A. Aleti, C. Trubiani, A. van Hoorn, P. Jamshidi, *An Efficient Method for Uncertainty Propagation in Robust Software Performance Estimation*, Elsevier Journal of Systems and Software (JSS), 2018. doi:10.1016/j.jss.2018.01.010 [SJR rating: **Q1**]
- ACM TOIT [J16] C. Pahl, P. Jamshidi, O. Zimmermann, *Architectural Principles for Cloud Software*, ACM Transactions on Internet Technology (TOIT), 18(2), 2018. doi:10.1145/3104028 [SJR rating: **Q1**]
- IEEE TCC [J15] C. Pahl, A. Brogi, J. Soldani, P. Jamshidi, *Cloud Container Technologies: a State-of-the-Art Review*, IEEE Transactions on Cloud Computing (TCC). 2017 doi:10.1109/TCC.2017.2702586. [SJR rating: **Q1**]
- WILEY JSEP [J14] C Pahl, P. Jamshidi, D Weyns, *Cloud architecture continuity: Change models and change rules for sustainable cloud software architectures*, Wiley Journal of Software: Evolution and Process (JSEP), 29(2), 2017 doi:10.1002/smr.1849. [SJR rating: **Q2**]
- ACM TAAS [J13] **Invited** A. Filieri, M. Maggio, K. Angelopoulos, N. D'Ippolito, I. Gerostathopoulos, A. Hempel, H. Hoffmann, P. Jamshidi, E. Kalyvianaki, C. Klein, F. Krikava, S. Misailovic, A. V. Papadopoulos, S. Ray, A. M. Sharifloo, S. Shevtsov, M. Ujma and T. Vogel, *Control Strategies for Self-Adaptive Software Systems*, ACM Transactions on Autonomous and Adaptive Systems (TAAS), invited paper, 11(4), 2017, doi:10.1145/3024188. [SJR rating: **Q1**]
- IEEE CLOUD [J12] P. Jamshidi, C. Pahl, N. Mendonça, *Managing Uncertainty in Autonomic Cloud Elasticity Controllers*, IEEE Cloud Computing, 2016. doi:10.1109/MCC.2016.66
- IEEE SOFTWARE [J11] **Featured Article** A. Balalaie, A. Heydarnoori, P. Jamshidi, *Microservices Enables DevOps: an Experience Report on Migration to a Cloud-Native Architecture*, IEEE Software, 2016. doi:10.1109/MS.2016.64, [SJR rating: **Q1**]
- IEEE TCC [J10] F. Fowley, C. Pahl, P. Jamshidi, D. Fang, X. Liu, *A Classification and Comparison Framework for Cloud Service Brokerage Architectures*, IEEE Transactions on Cloud Computing (TCC), 2016, doi:10.1109/TCC.2016.2537333. [SJR rating: **Q1**]
- WILEY SPE [J9] P. Jamshidi, C. Pahl, N. C. Mendonça, *Pattern-based Multi-Cloud Architecture Migration*, Wiley Software: Practice and Experience (SPE), 47(9), 1159-1184, 2016. doi:10.1002/spe.2442 [SJR rating: **Q2**]
- ★ ELSEVIER FGCS [J8] S. Farokhi, P. Jamshidi, E. B. Lakew, I. Brandic, E. Elmroth, *A Hybrid Cloud Controller for Vertical Memory Elasticity: A Control-theoretic Approach*, Elsevier Future Generation Computer Systems (FGCS), 65, 57 – 72 (2016). doi:10.1016/j.future.2016.05.028, [SJR rating: **Q1**]
- ELSEVIER FGCS [J7] D. Fang, X. Liu, I. Romdhani, P. Jamshidi, C. Pahl, *An Agility-Oriented and Fuzziness-Embedded Semantic Model for Collaborative Cloud Service Search, Retrieval and Recommendation*, Elsevier Future Generation Computer Systems (FGCS), 56, 11 – 26 (2016). doi:10.1016/j.future.2015.09.025, [SJR rating: **Q1**]

- IEEE TCC [J6] **Featured Article** P. Jamshidi, A. Ahmad, C. Pahl, *Cloud Migration Research: A Systematic Review*, IEEE Transactions on Cloud Computing (TCC), 1(2), 142 – 157 (2013).  doi:10.1109/TCC.2013.10, [SJR rating: **Q1**]
- SPRINGER JSEP [J5] A. Ahmad, P. Jamshidi, C. Pahl, *Classification and Comparison of Architecture Evolution Reuse Knowledge - A Systematic Review*, Springer Journal of Software: Evolution and Process (JSEP), 26(7): 654–691 (2014).  doi:10.1002/smr.1643, [SJR rating: **Q2**]
- EASST [J4] A. Ahmad, P. Jamshidi, C. Pahl, F. Khaliq, *A Pattern Language for the Evolution of Component-based Software Architectures*, Electronic Communications of the EASST, 59, 1 – 32 (2014).  doi:10.14279/tuj.eceasst.59.931
- IEEE SYSTEMS [J3] A. Khoshkbarforoushha, P. Jamshidi, M. Fahmideh, L. Wang, R. Ranjan, *Metrics for BPEL Process Reusability Analysis in a Workflow System*, IEEE Systems Journal, 1 – 10 (2014).  doi:10.1109/JSYST.2014.2317310, [SJR rating: **Q1**]
- SPRINGER SoSYM [J2] M. Fahmideh, M. Sharifi, P. Jamshidi, *Enhancing the OPEN Process Framework with Service-Oriented Method Fragments*, Springer Software and Systems Modeling (SoSym), 13(1): 361 – 390 (2014).  doi:10.1007/s10270-011-0222-z, [SJR rating: **Q1**]
- SPRINGER SOCA [J1] A. Khoshkbarforoushha, P. Jamshidi, A. Nikraves, F. Shams, *Metrics for BPEL process context-independency analysis*, Springer Service Oriented Computing and Applications (SOCA), 5(3): 139 – 157 (2011).  doi:10.1007/s11761-011-0077-8, [SJR rating: **Q2**]

REFEREED CONFERENCE PAPERS

- ★SoCC [C44] M.S. Iqbal, Z. Zhong, I. Ahmad, B. Ray, P. Jamshidi, *CAMEO: A Causal Transfer Learning Approach for Performance Optimization of Configurable Computer Systems.*, In Proc. of ACM Symposium on Cloud Computing (SoCC'23), 2023 [Acceptance rate: 29%(29/100); CORE rating: rank **A***].
- ADVML- FRONTIERS [C43] F. Ghofrani, M. Yaghouti, P. Jamshidi, *Rethinking Robust Contrastive Learning from the Adversarial Perspective*, In Proc. of the 2nd New Frontiers in Adversarial Machine Learning (AdvML Frontiers at ICML'23), 2023.
- ★EUROMLSYS [C42] M. Salmani, S. Ghafouri, A. Sanaee, K. Razavi, M. Mühlhäuser, J. Doyle, P. Jamshidi, M. Sharifi, *Reconciling High Accuracy, Cost-Efficiency, and Low Latency of Inference Serving Systems*, In Proc. of Proceedings of the 3rd Workshop on Machine Learning and Systems (EuroMLSys), 2023.
- CLOUD [C41] A. Mokhtari, A. Hossen, P. Jamshidi, M. Amini Salehi, *FELARE: Fair Scheduling of Machine Learning Applications on Heterogeneous Edge Systems*, In Proc. of International Conference on Cloud Computing (IEEE CLOUD), 2022 [Acceptance rate: 20%].
- ★AUTOML [C40] S. Iqbal, J. Su, L. Kotthoff, P. Jamshidi, *Getting the Best Bang For Your Buck: Choosing What to Evaluate for Faster Bayesian Optimization*, International Conference on Automated Machine Learning (AutoML), 2022.
- ★EuroSys [C39] S. Iqbal, R. Krishna, M.A. Javidian, B. Ray, P. Jamshidi, *Unicorn: Reasoning about Configurable System Performance through the lens of Causality*, In Proc. of European Conference on Computer Systems (EuroSys), 2022 [Acceptance rate: 25%(42/162); CORE rating: rank **A***].
- ★ARXIV [C38] Kimia Noorbakhsh, Modar Sulaiman, Mahdi Sharifi, Kallol Roy, P. Jamshidi, *Pretrained Language Models are Symbolic Mathematics Solvers too!*, [under review, available on arxiv].
- ★ICSE [C37] M. Velez, P. Jamshidi, N. Siegmund, S. Apel, C. Kaestner, *On Debugging the Performance of Configurable Software Systems: Developer Needs and Tailored Tool Support*, In Proc. of International Conference on Software Engineering (ICSE), 2022 [Acceptance rate: 26%(197/751); CORE rating: rank **A***].
- ★NEURIPS WHY-21 [C36] M.A. Javidian, O. Pandey, P. Jamshidi, *Scalable Causal Domain Adaptation*, In Proc. NeurIPS WHY-21 (Causal Inference and Machine Learning: Why now?), 2021 [Invited for oral presentation; Acceptance rate: 8%(4/50)].
- ★ICSE [C35] M. Velez, P. Jamshidi, N. Siegmund, S. Apel, C. Kaestner, *White-Box Analysis over Machine Learning: Modeling Performance of Configurable Systems*, In Proc. of International Conference on Software Engineering (ICSE), Virtual (May 2021) [Acceptance rate: 23%(138/602); CORE rating: rank **A***].
- AAMAS [C34] M. Rahman, A. Rasheed, M. Khan, M.A. Javidian, P. Jamshidi, Md. Mamun-Or-Rashid, *Accelerating Recursive Partition-Based Causal Structure Learning Using An Improved Structure Refinement Approach*, In Proc. of International Conference on Autonomous Agents and Multiagent Systems (AAMAS), Virtual (May 2021) [Acceptance rate: 25%(152/612); CORE rating: rank **A***].
- NEURIPS ML FOR SYSTEMS [C33] S. Iqbal, R. Krishna, M.A. Javidian, B. Ray, P. Jamshidi, *CADET: A Systematic Method For Debugging Misconfigurations using Counterfactual Reasoning*, In Proc. of NeurIPS ML for Systems, Virtual (December 2020).

- PROFES [C32] A. Banijamali, P. Kuvaja, M. Oivo, P. Jamshidi, *Kuksa*: Self-adaptive Microservices in Automotive Systems*, In Proc. of International Conference on Product-Focused Software Process Improvement (PROFES), Virtual (November 2020).
- ★UAI [C31] M. Javidian, M. Valtorta, P. Jamshidi, *Learning LWF Chain Graphs: A Markov Blanket Discovery Approach*, In Proc. of The Conference on Uncertainty in Artificial Intelligence (UAI), Toronto, Canada (August 2020) [Acceptance rate: 27%(142/515); CORE rating: rank **A***].
- SUM [C30] M. Javidian, M. Valtorta, P. Jamshidi, *Order-Independent Structure Learning of Multivariate Regression Chain Graphs*, In Proc. of International Conference on Scalable Uncertainty Management (SUM), Compiegne, France (December 2019).
- PROFES [C29] A. Banijamali, P. Jamshidi, P. Kuvaja, and M. Oivo, *Kuksa: A Cloud-Native Architecture for Enabling Continuous Delivery in the Automotive Domain*, In Proc. of International Conference on Product-Focused Software Process Improvement (PROFES), Barcelona, Spain (November 2019).
- ICGSE [C28] M. Viggiano, J. Oliveira, E. Figueiredo, P. Jamshidi, and C. Kaestner, *Understanding similarities and differences in software development practices across domains*, In Proc. of International Conference on Global Software Engineering (ICGSE), Montreal, Canada, (May 2019).
- ICPE [C27] C. Bezemer, S. Eismann, V. Ferme, J. Grohmann, R. Heinrich, P. Jamshidi, W. Shang, A. van Hoorn, M. Villavicencio, J. Walter, and F. Willnecker, *How is Performance Addressed in DevOps?*, In Proc. of International Conference on Performance Engineering (ICPE), Mumbai, India, (April 2019).
- OPML [C26] M. S. Iqbal, L. Kotthoff, P. Jamshidi, *Transfer Learning for Performance Modeling of Deep Neural Network Systems*, In Proc. of the USENIX Conference on Operational Machine Learning (OpML), Santa Clara, CA, (May 2019).
- ★SEAMS [C25] P. Jamshidi, J. Cam  ra, B. Schmerl, C. K  stner, D. Garlan, *Machine Learning Meets Quantitative Planning: Enabling Self-Adaptation in Autonomous Robots*, In Proc. of the 12th International Symposium on Software Engineering for Adaptive and Self-Managing Systems (SEAMS), Montreal, Canada, (May 2019) [Acceptance rate: 28%(10/36); CORE rating: rank **A**].
- AAAI WHY-19 [C24] M. A. Javidian, P. Jamshidi, M. Valtorta, *Transfer Learning for Performance Modeling of Configurable Systems: A Causal Analysis*, In Proc. of AAAI Spring Symposium Beyond Curve Fitting: Causation, Counterfactuals, and Imagination-based AI (WHY-19), Stanford, CA, USA, (March 2019).
- AAMAS [C23] M. A. Javidian, P. Jamshidi, R. Ramezani, *Avoiding Social Disappointment in Elections*, In Proc. of International Conference on Autonomous Agents and Multiagent Systems (AAMAS), Montreal, Canada, (May 2019). [CORE rating: rank **A***].
- FSE [C22] P. Jamshidi, M. Velez, C. K  stner, N. Siegmund, *Learning to Sample: Exploiting Similarities Across Environments to Learn Performance Models for Configurable Systems*, In Proc. of the ACM SIGSOFT Symposium on the Foundations of Software Engineering (FSE), Florida, USA, (Nov 2018) [Acceptance rate: 19%(55/295); CORE rating: rank **A***].
- TECHDEBT [C21] A. Mori, G. Vale, M. Viggiano, J. Oliveira, E. Figueiredo, E. Cirilo, P. Jamshidi, and C. K  stner, *Evaluating Domain-Specific Metric Thresholds: An Empirical Study*, In Proc. of the International Conference on Technical Debt (TechDebt), Gothenburg, Sweden, (May 27-28, 2018).

- ★ASE [C20] P. Jamshidi, N. Siegmund, M. Velez, C. Kästner, A. Patel, Y. Agarwal, *Transfer Learning for Performance Modeling of Configurable Systems: An Exploratory Analysis*, In Proc. of the 32nd IEEE/ACM International Conference on Automated Software Engineering (ASE), Illinois, USA, (Nov 2017) [Acceptance rate: 21%(67/322); CORE rating: rank **A***].
- ★SEAMS [C19] P. Jamshidi, M. Velez, C. Kästner, N. Siegmund, P. Kawthekar, *Transfer Learning for Improving Model Predictions in Highly Configurable Software*, In Proc. of the 12th International Symposium on Software Engineering for Adaptive and Self-Managing Systems (SEAMS), Buenos Aires, Argentina, (May 2017) [Acceptance rate: 23% (14/61), [Invited for an extension to ACM TAAS](#)]. CORE rating: rank **A**].
- CCGRID [C18] H. Arabnejad, C. Pahl, P. Jamshidi, G. Estrada, *A Comparison of Reinforcement Learning Techniques for Fuzzy Cloud Auto-Scaling*, in Proc. of The 17th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid), Madrid, Spain, (May 2017) [Acceptance rate: 23% (64/280); CORE rating: rank **A**]. **Nominated for the best paper award.**
- WICSA [C17] M. Bersani, F. Marconi, D. Tamburri, P. Jamshidi, A. Nodari, *Continuous Architecting of Stream-Based Systems*, In Proc. of The 13th Working IEEE/IFIP Conference on Software Architecture (WICSA), Venice, Italy, (April 2016). [Acceptance rate: 37% (56/149); CORE rating: rank **A**]
- ★MASCOTS [C16] P. Jamshidi, G. Casale, *An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems*, In Proc. of IEEE 24th International Symposium on Modeling, Analysis and Simulation of Computer and Telecommunication Systems (MASCOTS), London, UK (September 2016). [Acceptance rate: 17% (34/200); CORE rating: rank **A**]
- ★QoSA [C15] P. Jamshidi, A. Sharifloo, C. Pahl, H. Arabnejad, A. Metzger, G. Estrada, *Fuzzy Self-Learning Controllers for Elasticity Management in Dynamic Cloud Architectures*, In Proc. of 12th International ACM SIGSOFT Conference on the Quality of Software Architectures (QoSA), Venice, Italy, (April 2016). [CORE rating: rank **A**]
- ICCAC [C14] P. Jamshidi, A. Sharifloo, C. Pahl, A. Metzger, G. Estrada, *Self-Learning Cloud Controllers: Fuzzy Q-Learning for Knowledge Evolution*, In Proc. of IEEE International Conference on Cloud and Autonomic Computing (ICCAC), Boston, MA, USA, (Sept. 2015).
- ICAC [C13] Soodeh Farokhi, P. Jamshidi, D. Lucanin, I. Brandic, *Performance-Based Vertical Memory Elasticity*, In Proc. of IEEE International Conference on Autonomic Computing (ICAC), Grenoble, France, (Jul. 2015).
- ECSA [C12] C. Pahl, P. Jamshidi, *Software Architecture for the Cloud - A Roadmap Towards Control-Theoretic, Model-Based Cloud Architecture*, In Proc. of Springer European Conference on Software Architecture (ECSA), (Sept. 2015). [CORE rating: rank **A**]
- SEAMS [C11] A. Filieri, M. Maggio, K. Angelopoulos, N. D'Ippolito, I. Gerostathopoulos, A. Hempel, [Invited](#) H. Hoffmann, P. Jamshidi, E. Kalyvianaki, C. Klein, F. Krikava, S. Misailovic, A. V. Papadopoulos, S. Ray, A. M. Sharifloo, S. Shevtsov, M. Ujma and T. Vogel, *Software Engineering Meets Control Theory*, In Proc. of the 10th ACM International Symposium on Software Engineering for Adaptive and Self-Managing Systems, Firenze, Italy, (May 2015), [Acceptance rate: 29% (16/55); CORE rating: rank **A**]

- UCC [C10] L. Zhang, Y. Zhang, P. Jamshidi, L. Xu, C. Pahl, *Workload Patterns for Quality-Driven Dynamic Cloud Service Configuration and Auto-Scaling*, In Proc. of IEEE/ACM 7th International Conference on Utility and Cloud Computing (UCC), London, UK, (Dec 2014), [Acceptance rate: 19% (38/198); CORE rating: rank **A**]
- SEAMS [C9] P. Jamshidi, A. Ahmad, C. Pahl, *Autonomic Resource Provisioning for Cloud-Based Software*, In Proc. of the 9th ACM International Symposium on Software Engineering for Adaptive and Self-Managing Systems, Hyderabad, India, (Jun. 2014), [Acceptance rate= 18% (15/80); CORE rating: rank **A**].
- CSMR [C8] P. Jamshidi, M. Ghafari, A. Ahmad, C. Pahl, *A Framework for Classifying and Comparing Architecture-Centric Software Evolution Research*, In Proc. of 17th European Conference on Software Maintenance and Reengineering (CSMR), Genova, Italy, (Mar. 2013), [Acceptance rate: 36% (29/80)]
- CBSE [C7] M. Ghafari, P. Jamshidi, S. Shahbazi, H. Haghighi, *An architectural approach to ensure globally consistent dynamic reconfiguration of component-based systems*, In Proc. of the 15th ACM SIGSOFT symposium on Component-Based Software Engineering (CBSE), Bertinoro, Ital, (Sept. 2012). [Acceptance rate: 29%; CORE rating: rank **A**]
- CAiSE [C6] A. Ahmad, P. Jamshidi, C. Pahl, *Graph-Based Pattern Identification from Architecture Change Logs*, In Proc. of Springer International Conference on Advanced Information Systems Engineering (CAiSE), (Jun. 2012). [Short Paper, CORE rating: rank **A**]
- QSIC [C5] A. Kazemi, A. Rostampour, A. Zamiri, P. Jamshidi, H. Haghighi, F. Shams, *An Information Retrieval Based Approach for Measuring Service Conceptual Cohesion*, In Proc. of 11th IEEE International Conference on Quality Software (QSIC), Madrid, Spain, (Jul. 2011). [Acceptance rate: 17.6%]
- SCC [C4] A. Kazemi, A. Nasirzadeh, A. Rostampour, H. Haghighi, P. Jamshidi, F. Shams, *Measuring the Conceptual Coupling of Services Using Latent Semantic Indexing*, In Proc. of IEEE International Conference on Services Computing (SCC), Washington, DC, USA, (Jul. 2011). [Acceptance rate: 17%; CORE rating: rank **A**]
- SERVICES [C3] A. Kazemi, A. Rostampour, P. Jamshidi, E. Nazemi, F. Shams, A. Nasirzadeh, *A Genetic Algorithm Based Approach to Service Identification*, In Proc. of IEEE World Congress on Services (SERVICES), Washington, DC, USA, (Jul. 2011). [Acceptance rate: 17%; CORE rating: rank **A**]
- SERVICES [C2] A. Khoshkbarforousha, R. Tabein, P. Jamshidi, F. Shams, *Towards a metrics suite for measuring composite service granularity level appropriateness*, In Proc. of IEEE World Congress on Services (SERVICES), Miami, FL, USA, (Jul. 2010). [Acceptance rate: 18% (29/165)]
- SCC [C1] P. Jamshidi, M. Sharifi, S. Mansour, *To Establish Enterprise Service Model from Enterprise Business Model*, in Proc. of IEEE International Conference on Services Computing (SCC), Honolulu, HI, USA, (Jul. 2008). [Acceptance rate: 18%; CORE rating: rank **A**]

- RAP4ROBOTS [C44] H. Damirchi, F. Agostinelli, P. Jamshidi, *Independent Modular Networks*, In Proc. of Workshop on Effective Representations, Abstractions, and Priors for Robot Learning (RAP4Robots at ICRA'23), 2023.
- ARXIV [TR4] AM. Roth, N. Topin, P. Jamshidi, M. Veloso, *Conservative Q-improvement: Reinforcement learning for an interpretable decision-tree policy*, Arxiv, (2019).
- FC [TR3] S. Farokhi, P. Jamshidi, I. Brandic, E. Elmroth, *Self-adaptation Challenges for Cloud-based Applications: A Control Theoretic Perspective*, International Workshop on Feedback Computing, (2015).
- DAGSTUHL [TR2] A. van Hoorn, P. Jamshidi, P. Leitner, I. Weber, *Software Performance Engineering in the DevOps World*, Report from GI-Dagstuhl Seminar 16394, (Sept. 2017), <https://arxiv.org/abs/1709.08951>.
- SPEC [TR1] A. Brunnert, A. van Hoorn, F. Willnecker, A. Danciu, Wi. Hasselbring, C. Heger, N. Herbst, P. Jamshidi, R. Jung, J. von Kistowski, A. Koziolk, J. Kroß, S. Spinner, C. Vögele, J. Walter, A. Wert, *Performance-oriented DevOps: A Research Agenda*, SPEC Research Group — DevOps Performance Working Group, Standard Performance Evaluation Corporation (SPEC), (Aug. 2015), SPEC-RG-2015-01.

SOFTWARE ARTIFACTS

- Github Almost all listed software is developed collaboratively (LD: lead developer; CC: contributor). AISys GitHub Organization; My personal GitHub Account; We also actively maintain several GitHub Organizations associated with our major projects, please find the links on my homepage.
- CC [S17] **Unicorn**, *Unicorn is a framework for reasoning about system performance using Causal AI*, Python,  <https://github.com/softsys4ai/unicorn>
- ★CC [S17] **ATHENA**, *is a Framework for defending machine learning systems against adversarial attacks*, Python,  <https://github.com/softsys4ai/athena>
- LD [S16] **robot_control**, *robot_control is a set of controllers and actuators that run, control, and interface the ROS-enabled robots, as a part of DARPA BRASS project.*, C++,  https://github.com/pooyanjamshidi/robot_control
- LD [S15] **brass_gazebo_battery**, *brass_gazebo_battery is a Gazebo plugin that simulates an open-circuit battery model. This is a fairly extensible and reusable battery plugin for any kind of Gazebo-compatible robots.*, C++,  https://github.com/pooyanjamshidi/brass_gazebo_battery
- LD [S14] **GenPerf**, *GenPerf uses symbolic regression to synthetically generate target performance influence models with different similarities to the source model. GenPerf is used to generate synthetic data for evaluating our TL approach.*, Python,  <https://github.com/pooyanjamshidi/GenPerf>
- LD [S13] **AutoTL**, *This tool enables an adaptive sampling that learns from multiple exclusive information origins including influential configuration options, their interactions, and performance distribution of the configurable software*, Python,  <https://github.com/pooyanjamshidi/autotl>
- LD [S12] **model-learner**, *This tool enables discovering a black box model using regression models and transfer learning. This was used in the BRASS project to enable battery charge/recharge in a self-adaptive loop*, Python,  <https://github.com/cmu-mars/model-learner>
- LD [S11] **autoscaling-bigdata**, *A library for application level runtime monitoring and runtime change actuators and auto-scaling controllers for Big Data technologies such as Apache Storm, Spark, Hadoop, Matlab+REST APIs*,  <https://github.com/pooyanjamshidi/autoscaling-bigdata>
- LD [S10] **TL4CO**, *A Machine Learning tool for finding the optimum configuration of Big Data systems by transferring the learning from other system versions in DevOps context*, Matlab+Java,  <https://github.com/dice-project/DICE-Configuration-TL4CO>
- ★LD [S9] **BO4CO**, *A Machine Learning tool for finding the optimum configuration of Big Data systems*, Matlab+Python,  <https://github.com/dice-project/DICE-Configuration-BO4CO>
- LD [S8] **ElasticBench**, *A cloud application framework to plug-in auto-scaling logic and experimentally evaluate controllers in a feedback control loop on platform as a service environment on Microsoft Azure*, .NET,  <https://github.com/pooyanjamshidi/ElasticBench>
- CC [S7] **spark-suite**, *A suite for automated configuration testing, automated topology deployment and a benchmarking tool for Apache Spark*, Java,  <https://github.com/pooyanjamshidi/spark-suite>

- CC [S6] **OSTIA**, *A parser to elicit and represent Storm topologies by reverse engineering Storm-based programs*, Ruby,  <https://github.com/maelstromdat/OSTIA>
- LD [S5] **pong-engine**, *An engine that runs pong games on Matlab and paddles are controlled by reinforcement learner agents. I implemented this piece of software for a reinforcement learning course*, Matlab,  <https://github.com/pooyanjamshidi/pong-engine>
- CC [S4] **MDLoad**, *MDload is a model-driven workload generation tool that automatically generates requests to a web application by simulating a set of users*, Java+Matlab,  <https://github.com/imperial-modacLOUDS?query=modacLOUDS-mdload>
- LD [S3] **Fuzzy-Q-Learning**, *An implementation of Fuzzy Q-Learning for making cloud auto-scaling more intelligent through online policy learning*, Matlab,  <https://github.com/pooyanjamshidi/Fuzzy-Q-Learning>
- LD [S2] **RobusT2Scale**, *A cloud auto-scaler based on fuzzy reasoning*, Matlab,  <https://github.com/pooyanjamshidi/RobusT2Scale>
- LD [S1] **ASIM**, *A program that automatically identifies services out of business processes*, Java,  <https://github.com/pooyanjamshidi/ASIM>

DATA

I release the data that I collect for my research to the public community for replication.

- [D3] **ATHENA**, *The dataset (30GB) associated with ATHENA—a framework based on Diverse Weak Defenses for building Adversarial Defense.*,  <https://zenodo.org/record/4141383>
- [D2] **ASE 2017**, *Transfer Learning for Performance Modeling of Configurable Systems: An Exploratory Analysis*, **Subject systems:** SaC, SQLite, SPEAR, X264,  <https://github.com/pooyanjamshidi/ase17>
- [D1] **MASCOTS 2016**, *An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems*, **Subject systems:** Apache Storm, Apache Spark, Apache Hadoop, Apache Cassandra,  <https://zenodo.org/record/56238>

FUNDING

- NSF EAGER **“Collaborative Research: EAGER: Towards a Design Methodology for Software-Driven Sustainability”**, Agency: *NSF*, Award Amount: **\$300,000**, Project Period: 09/01/2022 - 08/31/2023
Role: Co-PI. PI: Eunsuk Kang (Carnegie Mellon University); other co-PIs: Mehdi Mirakhorli and Callie Babbitt (Rochester Institute of Technology).
- NSF MEDIUM **“Collaborative Research: SHF: Medium: Causal Performance Debugging for Highly-Configurable Systems”**, Agency: *NSF*, Award Amount: **\$1,200,000**, Project Period: 10/01/2021 - 09/30/2024
Role: PI. Co-PIs: Christian Kaestner (CMU) and Baishakhi Ray (Columbia).
- NSF LARGE **“RTG: Mathematical Foundation of Data Science at University of South Carolina”**, Agency: *NSF*, Award Amount: **\$1,996,609**, Project Period: 08/01/2021 - 07/31/2026
Role: Co-PI. Co-PIs (colleagues in the Math department): Wolfgang Dahmen, Linyuan Lu (PI), Wuchen Li, and Qi Wang.
- NSF SMALL **“SmartSight: an AI-Based Computing Platform to Assist Blind and Visually Impaired People”**, Agency: *NSF*, Award Amount: **\$499,650**, Project Period: 10/01/2020 - 10/01/2023
Role: PI. Co-PIs: Mohsen Amini Salehi (UL).
- NASA **“RASPBERRY SI: Resource Adaptive Software Purpose-Built for Extraordinary Robotic Research Yields - Science Instruments”**, Agency: *NASA*, Award Amount: **\$300,000**, Project Period: 10/01/2020 - 10/01/2023
Role: PI. Co-PIs: David Garlan (CMU, co-I), Bradley Schmerl (CMU, co-I), Javier Camara (York, collaborator), Ellen Czaplinski, Katherine Dzurilla (University of Arkansas, consultant), Matt DeMinico (NASA Glenn Research Center, consultant), Mike Dalal (NASA Ames, testbed), Hari Nayar (NASA JPL, testbed).
- NASA **“A Generic Data-Driven Framework via Physics-Informed Deep Learning”**, Agency: *NASA*, Award Amount: **\$100,000**, Project Period: 08/01/2020 - 08/01/2021
Role: Co-PI. PI: Lang Yuan (Mechanical Engineering).
- NASA **“Robust Software Testing of Autonomous Aerospace Robotic Systems Using Transfer Learning”**, Agency: *NASA*, Award Amount: **\$50,000** (25,000 cost share), Project Period: 05/07/2019 - 05/06/2020
Role: PI; Co-PIs: Gregory Gay, Jason O’Kane.
- DARPA-DOD-AFOSR (SUB-AWARD) **“Online Transfer Learning and Self-Adaptation of Robots”**, Agency: *Air Force Office of Science Research (AFOSR) and Defense Advanced Research Projects Agency (DARPA)*, Award Amount: **\$114,741** (sub-award only), Project Period: 09/01/2018 - 08/31/2019
Role: PI (subaward); Co-PIs: Jonathan Aldrich (PI), David Garlan, Christian Kaestner, Claire Le Goues, and Manuela Veloso.
- ROBLOX **“Unrestricted Gift”**, Agency: *Roblox Corporation*, Award Amount: **\$20,000**, Fund Approved: May 2023
Role: PI. Collaborator: Tania Lorido-Botran (Roblox).
- GOOGLE **“GCP research credits for Adversarial ML”**, Agency: *Google*, Award Amount: **\$20,000**, Project Period: 09/01/2019 - 06/01/2020
Role: PI.

BIG DATA HEALTH SCIENCE CENTER (INTERNAL) **“Identifying Optimal Vaccine Promotion Messages for Vulnerable Subgroups from Large-Scale Gamified Interventions”**, Agency: *University of South Carolina*, Award Amount: **\$30,000**, Project Period: 08/16/2021 - 08/16/2022
Role: Co-PI, PI: Gregory Trevors

ASPIRE-I (INTERNAL) **“Optimizing Energy Consumption of Deep Neural Networks for Intelligent Learning Systems”**, Agency: *University of South Carolina ASPIRE-I*, Award Amount: **\$15,000**, Project Period: 07/01/2019 - 09/30/2020
Role: PI

MAGELLAN SCHOLAR AWARD (INTERNAL) **“Multi-stage Compression of Deep Neural Networks through Pruning and Knowledge Distillation”**, Agency: *UofSC Office of Undergraduate Research*, Award Amount: **\$2,750**, Project Period: 01/01/2020 - 12/31/2020
Role: PI.

MAGELLAN SCHOLAR AWARD (INTERNAL) **“Ensemble of Many Weak Defenses: Defending Deep Neural Networks Against Adversarial Attacks”**, Agency: *UofSC Office of Undergraduate Research*, Award Amount: **\$2,750**, Project Period: 01/01/2020 - 12/31/2020
Role: PI.

SURF (INTERNAL) **“Adversarial Machine Learning”**, Agency: *USC Honors College SURF Grant*, Award Amount: **\$2,000**, Project Period: 10/15/2019 - 06/30/2020
Role: PI.

MCNAIR (INTERNAL) **“Bayesian Structure Learning”**, Agency: *University of South Carolina McNair Junior Fellows*, Award Amount: **\$3,000**, Project Period: 05/15/2019 - 08/16/2019
Role: PI.

SURF (INTERNAL) **“Neurofeedback-based Reinforcement Learning”**, Agency: *USC Honors College SURF Grant*, Award Amount: **\$2,600**, Project Period: 10/15/2018 - 06/30/2019
Role: PI.

MEDIA COVERAGE AND PRESS RELEASE

Breakthrough interview	An interview with the University of South Carolina's breakthrough magazine: https://sc.edu/uofsc/posts/2022/06/breakthrough_star_pooyan_jamshidi.php
Podcast Interview	Dr. Alireza Dehghani, in his recent podcast, interviews me about my journey from undergraduate to becoming an assistant professor at UofSC, August 2022, https://www.youtube.com/watch?v=ZZarpnDK0Y0
Interview with Bungee Pi Part 2	Learn math if you want to get into AI, June 2022, https://www.youtube.com/watch?v=-Fkv180in8I
Interview with Bungee Pi Part 1	Breakthrough Ideas in Artificial Intelligence and Mathematics, June 2022, https://www.youtube.com/watch?v=YdwQXW2z1bs
Causal AI for Systems	Research intends to develop a shift in testing and debugging for modern machine learning systems, January 2022, https://www.sc.edu/study/colleges_schools/engineering_and_computing/news_events/news/2022/research_intends_to_develop_a_shift_in_testing_and_debugging_for_modern_machine_learning_systems.php
AI for Social Good	A new way to 'see' , June 2021, https://www.sc.edu/uofsc/posts/2021/06/smart_sight.php
AI in Space	USC researcher wants to train robots for NASA deep space missions, December 2020, https://www.postandcourier.com/columbia/news/usc-researcher-wants-to-train-robots-for-nasa-deep-space-missions/article_93d9bb3c-3afa-11eb-bd4c-7700ac496485.html
AI in Space	UofSC to develop AI-based autonomous systems for space missions, November 2020, https://www.sc.edu/study/colleges_schools/engineering_and_computing/news_events/news/2020/jamshidi_ai_space.php
AISys Lab	Media coverage of the AISys lab, June 2019, https://www.sc.edu/study/colleges_schools/engineering_and_computing/about/news/2019/jamshidiai.php

INVITED TALKS, LECTURES AND SEMINARS

TALKS	The slides of my recent talks are available at: https://pooyanjamshidi.github.io/talks/
ICSE/SEAMS'23	Learning from Valerie Issarny: Insights Gained from Program Co-Chairing SEAMS'23 , <i>Melbourne</i> , 2023
OKTOBERBEST	Experiential Learning by Building Real-World AI Systems , <i>Oktoberbest: A Symposium On Teaching</i> , 2022
NASA JPL	Understanding and Explaining the Root Causes of Performance Faults with Causal AI: A Path towards Building Dependable Computer Systems , <i>NASA JPL, Pasadena, CA</i> , 2022
EXPLORECSR	Causal AI for Systems , <i>exploreCSR: Democratizing AI</i> , 2022
IEEE SMDS	Causal AI for Systems , <i>Virtual</i> , 2021
STANFORD MLSYS	Causal AI for Systems , <i>Virtual</i> , 2021
DAGSTUHL SEMINAR	Causal Inference for Performance Analyses and Debugging of Serverless Systems , <i>Virtual</i> , 2021
ASE TUTORIAL	Machine Learning meets Software Performance , 2021
UNIVERSITY OF LOUISIANA	ATHENA: A Framework based on Diverse Weak Defenses for Building Adversarial Defense , <i>Virtual</i> , November 2020
CMU	ATHENA: A Framework based on Diverse Weak Defenses for Building Adversarial Defense , <i>Virtual</i> , September 2020
UPENN	ATHENA: A Framework based on Diverse Weak Defenses for Building Adversarial Defense , <i>Virtual</i> , September 2020
GOOGLE HEADQUARTERS	Ensembles of Many Diverse Weak Defenses can be Strong: Defending Deep Neural Networks Against Adversarial Attacks , <i>Virtual</i> , September 2020
AI SUMMIT AT SILICON VALLEY	Robust Causal Transfer Learning , <i>Virtual</i> , September 2020
AUGUSTA UNIVERSITY	Ensembles of Many Diverse Weak Defenses can be Strong: Defending Deep Neural Networks Against Adversarial Attacks , <i>Augusta, Georgia</i> , February 2020
FURMAN UNIVERSITY	Transfer Learning for Performance Analysis of Machine Learning Systems , <i>Greenville, SC</i> , April 2019
SC EPSCoR CONFERENCE	Transfer Learning for Performance Analysis of Machine Learning Systems , <i>Greenville, SC</i> , April 2019
RE-WORK DEVOPS SUMMIT	Machine Learning meets DevOps: Transfer Learning for Performance Optimization , <i>Houston, Texas</i> , November 2018
SATURN	Architectural Tradeoffs in Learning-Based Software , <i>Plano, Texas</i> , May 2018
NC STATE UNIVERSITY	Learning Software Performance Models for Dynamic and Uncertain Environments , <i>Raleigh, US</i> , 2017
SPEC DEVOPS RG	An Exploratory Analysis of Transfer Learning for Performance Modeling of Configurable Systems , <i>Online talk, RG DevOps Performance Working Group</i> , 2017

- DAGSTUHL SEMINAR **Machine Learning meets DevOps**, *Software Performance Engineering in the DevOps World, Dagstuhl, Germany*, 2016
- BERN UNIVERSITY **An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems**, *Bern, Switzerland*, 2016
- SPEC DEVOPS RG **An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems**, *Online talk, RG DevOps Performance Working Group*, 2016
- SPEC DEVOPS RG **Microservices Architecture Enables DevOps: Migration to a Cloud-Native Architecture**, *Online talk, RG DevOps Performance Working Group*, 2016
- NC4 CONFERENCE **DevOps: Migration to a Cloud-Native Architecture**, *The National Conference on Cloud Computing & Commerce, Dublin, Ireland*, 2016
- SHARIF UNIVERSITY **Fuzzy Self-Learning Controllers for Elasticity Management in Dynamic Cloud Architectures**, *Tehran, Iran*, 2016
- NII SHONAN MEETING **Fuzzy Self-Learning Controllers for Elasticity Management in Dynamic Cloud Architectures**, *National Institute of Informatics (NII), Controlled Adaptation of Self-adaptive Systems (CASaS), Shonan, Japan*, 2016
- TRINITY COLLEGE **Self-learning Cloud Controllers**, *Trinity College Dublin*, 2015
- UFC UNIVERSITY **Self-learning Cloud Controllers**, *Federal University of Ceará, Fortaleza, Brazil*, 2015
- UECE UNIVERSITY **Cloud Migration Patterns: A Multi-Cloud Architectural Perspective**, *Ceará State University, Fortaleza, Brazil*, 2015
- NC4 CONFERENCE **Fuzzy Q-Learning for Knowledge Evolution**, *The National Conference on Cloud Computing & Commerce, Dublin, Ireland*, 2015
- SPEC DEVOPS RG **Self-learning Cloud Controllers**, *Online talk, RG DevOps Performance Working Group*, 2015
- DAGSTUHL SEMINAR **Fuzzy Control Meets Software Engineering**, *Control Theory meets Software Engineering, Dagstuhl, Germany*, 2014

TEACHING

CURRENT TEACHING

- Fall 2021-* **CSCE 212: Introduction to Computer Architecture**, *University of South Carolina, Columbia, SC*, Instructor, <https://pooyanjamshidi.github.io/csce212/>
- Fall 2018-* **CSCE 585: Machine Learning Systems**, *University of South Carolina, Columbia, SC*, Instructor, <https://pooyanjamshidi.github.io/mls/>
- Spring 2019-* **CSCE 580: Artificial Intelligence**, *University of South Carolina, Columbia, SC*, Instructor, <https://pooyanjamshidi.github.io/csce580/>

PAST TEACHING

- 4/2018 **S17-655 Architectures for Software Systems (CMU Software Engineering Masters Program)**, *Carnegie Mellon University, Pittsburgh, US*, A guest lecture on Machine Learning for the Software Architect
- 11/2016 **SMA: Software Modeling and Analysis (Oscar Nierstrasz's course)**, *Bern University, Switzerland*, A guest lecture on Architecture Extraction
- 10/2015–1/2016 **424H - Learning in Autonomous Systems**, *Imperial College London, TA*
- 11/2014–12/2014 **CA674 - Cloud Architecture**, *Dublin City University*, Lectures shared with Claus Pahl
- 3/2014–6/2014 **CA668 E-commerce Infrastructure**, *Dublin City University*, Lectures shared with Claus Pahl
- 10/2017 **Foundations of Software Engineering (Christian Kästner and Claire Le Goues)**, *Carnegie Mellon University*, Guest lecture on *Microservices*
- 9/2008-6/2010 **Software Engineering**, *Tarbiat Moallem University*, Lecturer
- 9/2001-6/2003 **Introduction to C/C++**, *Amirkabir University of Technology, TA*
- 9/2002-6/2003 **Data Structures**, *Amirkabir University of Technology, TA*

MENTORSHIP

POSTDOCS

2022- **Sonam Kharde**
2022- **Mehdi Yaghouti**
2019-2022 **Mohammad Ali Javidian**

DOCTORAL STUDENTS

2018- **Shahriar Iqbal**, *Defended his proposal in January'23*
2023- **Abdolrasool Sharifi**, *Transferred to USC from the University of Virginia*
2021- **Abir Hossen**
2021- **Fatehmeh Ghofrani**
2021- **Hamed Damirchi**
2019-2022 **Jianhai Su**
2019-2022 **Ying Meng**
2022-2023 **Nasrin Imanpour**
2020-2021 **Shuge Lei**

PHD INTERNS

2022 **Saeid Ghafouri**, *Queen Mary University of London*

UNDERGRADUATE STUDENTS (CURRENT)

4/2023- **Kartik Singhal**, *IIIT, Delhi, India*
1/2022- **Samuel Whidden**

MASTERS STUDENTS (COMPLETED)

2019-2020 **Peter Mourfield**

UNDERGRADUATE STUDENTS (COMPLETED)

9/2020-4/2023 **Kimia Noorbakhsh**, *Now: PhD student at MIT*
3/2022-12/2022 **Hung-Tien Huang**, *Now: PhD student at UNC Chapel Hill*
4/2020-6/2022 **Om Pandey**, *Now: Graduate student at Texas Austin*
4/2021-2/2022 **Ahana Biswas**, *Now: Graduate student at Pitt*
5/2021-8/2021 **Madelyn Khoury**, *REU*
5/2021-8/2021 **Bruce Brasseur**, *REU*
9/2020-4/2021 **Cody Shearer**, *RA*, *Now: Software Developer at Krumware*
8/2018-6/2019 **Nathan Stofik**, *RA*
12/2018-8/2019 **Tristan Klintworth**, *RA*
8/2019-5/2020 **Blake Edwards**, *RA*, *Now: AI Technologist at Boeing Research & Technology*
8/2019-5/2020 **Stephen Baione**, *RA*, *Now: Systems Design Engineer at AMD*
5/2019-8/2019 **Joshua Ravishankar**, *REU*
5/2019-8/2019 **Rabina Phuyel**, *REU*

PRE-COLLEGE INTERNS

- 6/2021- **Lane Stanley**, *Heathwood Hall High School, Columbia, SC*
2/2021- **Rohan Bafna**, *Riverside High School, Greenville, SC*
4/2021- **Rohit Rajagopalan**, *Southside High school, Greenville, SC*

RESEARCH CO-MENTORING

- 2023 **Carnegie Mellon University**, *REU Program*, Haesue Baik (first-year undergrad at University of Michigan, Ann Arbor), co-advised by Rohan Padhye, Vasudev Vikram (CMU PhD Student)
Understanding coverage guided fuzzing, understanding the performance of configurable systems, and designing custom fuzzing functions.
- 2022 **Carnegie Mellon University**, *REU Program*, Janhvi Somaiya (first-year undergrad at Rice University), co-advised by Christian Kaestner (CMU), Sonam Kharde (USC postdoc), Shahriar Iqbal (USC PhD student)
On identifying configuration-specific performance bugs with fuzzing.
- 2023 **Federal University of Minas Gerais (UFMG)**, *M.Sc. thesis*, Markos Vigiato de Almeida
On the Investigation of Software Development and Evolution Practices.
- 2017-2018 **Federal University of Minas Gerais (UFMG)**, *M.Sc. thesis*, Markos Vigiato de Almeida
On the Investigation of Software Development and Evolution Practices.
- 2018 **Carnegie Mellon University**, *Undergraduate research project*, Students: Alex Gao, Connor Lin, Jason Bak, Sander Lanbo Shi, Yunjie Su
Design space explorations of deep neural network architectures for embedded devices.
- 2017 **Carnegie Mellon University**, *REU Program*, Changming Xu, co-advised by Christian Kaestner (CMU)
Can you fool a self-adaptive software system?
- 2016-17 **Imperial College London**, *B.Eng. project*, Ka Yan Wong
Experimental study of performance variations in big data systems.
- 2016 **Imperial College London**, *M.Eng. project*, Yifan Zhai
A DevOps canary testbed for Big Data application testing.
- 2015 **Imperial College London**, *B.Eng. final project*, Zhang Haoran, Qiu Jiaxin, Abdeljallal Fahd, Lu Cong, Chadjiminis Ioannis, Liu Yao
A suite for automated configuration testing and benchmarking for Apache Spark.
- 2016 **Imperial College London**, *M.Eng. final project*, Xidi Chen
A suite for automated configuration testing and benchmarking for Apache Hadoop.
- 2014-2015 **Dublin City University**, *M.Sc. practicum*, Robert Mason
Auto-scaling in OpenStack cloud.
- 2014 **Dublin City University**, *MS.c. practicum*, Brian C. Carroll
Auto-scaling in the cloud: evaluating a control-based technique.
- 2014-15 **Sharif University of Technology**, *M.Sc. thesis*, Armin Balalaie
Migrating to cloud-native architectures using microservices.
- 2008-10 **Shahid Beheshti University**, *M.Sc. thesis*, Ali Rostampour, Ali Kazemi
A metric for measuring the degree of entity-centric service cohesion.

PHD COMMITTEE MEMBER

- 2023 **Mohammed E. Elbity**, *Advisor: Ramtin Zand*, University of South Carolina, Computer Science and Engineering
Approximate Computing and In-Memory Computing: The best of the two worlds!
- 2023 **Bharat Joshi**, *Advisor: Ioannis Rekleitis*, University of South Carolina, Computer Science and Engineering
Robust Underwater State Estimation and Mapping
- 2023 **Luc Lesoil**, *Advisor: Mathieu Acher*, INSA Rennes, France, Deep Software Variability for Resilient Performance Models of Configurable Systems
- 2022 **Manas Gaur**, *Advisor: Amit Sheth*, University of South Carolina, Computer Science and Engineering
Knowledge-Infused Learning
- 2022 **Aron Hein**, *Advisor: Homayoun Valafar*, University of South Carolina, Computer Science and Engineering
- 2022 **Olajide H. Bamidele**, *Advisor: Andreas Heyden*, University of South Carolina, Chemical Engineering
- 2021 **Rasika Jayarathna**, *Advisor: Jochen Lauterbach*, University of South Carolina, Chemical Engineering
- 2021 **Alireza Salahirad**, *Advisor: Greg Gay*, University of South Carolina, Computer Science and Engineering
Empirical Studies on Automated Software Testing Practices
- 2020 **Elizabeth Stewart**, *Advisor: Brett Sherman*, University of South Carolina, Philosophy
Alexa, Should I Trust You? A Theory of Trustworthiness for Artificial Intelligence
- 2020 **Nare Karapetyan**, *Advisor: Ioannis Rekleitis*, University of South Carolina, Computer Science and Engineering
Area Coverage Path Planning Problem in Aquatic Environments
- 2020 **Hazhar Rahmani**, *Advisor: Jason O’Kane*, University of South Carolina, Computer Science and Engineering
Automata theoretic approaches to planning in robotics: combinatorial filter minimization, planning to chronicle, and temporal logic planning with soft specifications
- 2020 **William Hoskins**, *Advisor:*, University of South Carolina, Computer Science and Engineering
- 2019 **Mohammad Ali Javidian**, *Advisor: Marco Valtorta*, University of South Carolina, Computer Science and Engineering
Advanced Topics in Probabilistic Graphical Models: Properties, Learning Algorithms, and Applications
- 2019 **Hussein Almulla**, *Advisor: Greg Gay*, University of South Carolina, Computer Science and Engineering
Learning How to Search: Generating Effective Test Cases Through Adaptive Fitness Function Selection
- 2019 **Shervin Ghasemlou**, *Advisor: Jason O’Kane*, University of South Carolina, Computer Science and Engineering
Algorithmic Robot Design: Label Maps, Procrustean Graphs, and the Boundary of Non-destructiveness

- 2018 **Hayder Dawood Abboud**, *Advisor: Andrea Benigni*, University of South Carolina, Electrical Engineering
Data-Driven Modeling Through Power Hardware in the Loop Experiments: A PV Micro-Inverter Example
- 2018 **Jason Moulton**, *Advisor: Ioannis Rekleitis*, University of South Carolina, Computer Science and Engineering
A Novel and Inexpensive Solution to Build Autonomous Surface Vehicles Capable of Negotiating Highly Disturbed Environments

M.S. COMMITTEE MEMBER

- 2020 **Noah Geveke**, *Advisor: Marco Valtorta*, University of South Carolina, Computer Science and Engineering
On the Robustness of Bayesian Network Learning Algorithms against Malicious Attacks

SERVICE

INTERNAL SERVICE AT THE UNIVERSITY OF SOUTH CAROLINA

- Department Graduate Committee, 2018–*
- Department Faculty Search Committee, 2019–2020
- University SPIRE grant review committee, 2019–2020
- University Magellan review committee, 2020–2021
- University Carolina Scholar mentor, 2019–*
- University Mentor for underrepresented minority students in an R25 NIH/NIAID funded program, total award:\$1.565m, 06/23-05/28

PROFESSIONAL SERVICE

CO-FOUNDING INITIATIVES

- JSys **Co-founded JSys (Journal of Systems Research)**, a new diamond open-access journal with Vijay Chidambaram, Neeraja Yadwadkar, Ivo Jimenez, and Romain Jacob,  <https://www.jsys.org/>
- Gamecock Robotics **Mentoring the Gamecock Robotics team**, a student-led team for a competitive robotics league called Vex Robotics,  <http://gamecockrobotics.github.io/>
- SPEC RG **Elected as the vice chair of SPEC RG Predictive Data Analytics**,  <https://research.spec.org/working-groups/rg-predictive-data-analytics/>

EDITORIAL

- JSys **Co-Editor at JSys (Journal of Systems Research)**, Two Areas: Computer Architecture (with Devashree Tripathy at IIT Bhubaneswar) and Configuration Management for Systems (with Tianyin Xu at UIUC),  <https://www.jsys.org/board>
- IEEE Software **IEEE Software Special Issue on Microservices, Guest editor**, Co-edited with James Lewis (ThoughtWorks), Stefan Tilkov (innoQ), Claus Pahl, and Nabor Mendonça, This special issue attracted 26 submissions, a record number in IEEE Software,  <https://www.computer.org/software-magazine/2017/02/10/microservices-call-for-papers/>

PROGRAM CO-CHAIR

- SEAMS 2023 The International Symposium on Software Engineering for Adaptive and Self-Managing Systems, Program Co-Chair with Valerie Issarny (update: Program Co-Chair with Radu Calinescu and Raffaella Mirandola).

AREA CHAIR

- AutoML-Conf 2023 2nd International Conference on Automated Machine Learning.
- AutoML-Conf 2022 1st International Conference on Automated Machine Learning.

CO-ORGANIZER

- SEAMS 2023 The International Symposium on Software Engineering for Adaptive and Self-Managing Systems, Program Co-Chair.

MLArchSys at ISCA 2023	ML for Computer Architecture and Systems, Co-Organizer and PC.
SEAMS 2022	The International Symposium on Software Engineering for Adaptive and Self-Managing Systems, social media chair.
ENSEMBLE 2019	2nd International Workshop on Ensemble-based Software Engineering for Modern Computing Platforms (co-located with ESEC/FSE 2019), co-chair.
SEAMS 2017	The International Symposium on Software Engineering for Adaptive and Self-Managing Systems, publicity and proceedings chair (Co-organized with David Garlan, Bashar Nuseibeh, Javier Camára, and Nicolás D'Ippolito).
CloudWays 2017	International Workshop on Cloud Adoption and Migration, Workshop co-chair (Co-organized with Claus Pahl, and Nabor Mendonça).
Dagstuhl 2016	Software Performance Engineering in the DevOps World, Co-organized the seminar with Andre van Hoorn, Philipp Leitner, and Ingo Weber.
CloudWays 2016	International Workshop on Cloud Adoption and Migration, Workshop co-chair.
CloudWays 2015	International Workshop on Cloud Adoption and Migration, Workshop co-chair.

PROGRAM COMMITTEES (CONFERENCES)

ICPE 2024	International Conference on Performance Engineering - Data Challenge
ICSE 2024	International Conference on Software Engineering
ICPE 2023	International Conference on Performance Engineering
ASE 2022	International Conference on Automated Software Engineering - Artifacts Track
SEAMS 2022	Software Engineering for Adaptive and Self-Managing Systems
ICPE 2022	International Conference on Performance Engineering
ICSE/NIER 2022	International Conference on Software Engineering
ASE 2021	International Conference on Automated Software Engineering - Artifacts Track
AISTATS 2021	International Conference on Artificial Intelligence and Statistics
SEAMS 2021	16th Software Engineering for Adaptive and Self-Managing Systems
ICSA 2021	International Conference on Software Architecture
ICPE 2021	International Conference on Performance Engineering
XP 2020	International Conference on Agile Software Development
ICSE 2020	International Conference on Software Engineering
SEAMS 2020	15th Software Engineering for Adaptive and Self-Managing Systems
ICPE 2020	International Conference on Performance Engineering
VaMoS 2020	International Conference on Variability Modeling of Software-Intensive Systems
ECSA 2019	13th European Conference on Software Architecture
SATURN 2019	SEI Architecture User Network (SATURN) Conference
ICSA 2019	International Conference on Software Architecture (Tool Track)
Microservices 2019	International Conference on Microservices
SEAMS 2019	14th Software Engineering for Adaptive and Self-Managing Systems
ICSOC 2018	16th International Conference on Service-Oriented Computing
ECSA 2018	12th European Conference on Software Architecture
ICDCS 2018	38th International Conference on Distributed Computing Systems
SEAMS 2018	13th Software Engineering for Adaptive and Self-Managing Systems

ECSA 2017 11th European Conference on Software Architecture
 SEAMS 2017 12th Software Engineering for Adaptive and Self-Managing Systems
 EUSPN 2017 8th Emerging Ubiquitous Systems and Pervasive Networks
 SIGMOD 2016 ACM SIGMOD 2016 Reproducibility
 SEAMS 2016 11th Software Engineering for Adaptive and Self-Managing Systems
 EUSPN 2016 7th Emerging Ubiquitous Systems and Pervasive Networks
 ICSOFT 2016 13th International Conference on Software Technologies
 ICSOFT 2015 12th International Conference on Software Technologies

PROGRAM COMMITTEES (WORKSHOPS)

WAM 2023 1st International Workshop on Application Modernization
 ASSYST 2023 Architecture and System Support for Transformer Models
 AMP 2021 International Workshop on Agility with Microservices Programming
 AKSAS 2018 International Workshop on Architectural Knowledge for Self-Adaptive Systems
 MLMH 2018 KDD Workshop on Machine Learning for Medicine and Healthcare
 AMS 2018 International Workshop on Architecting with MicroServices
 SQUADE 2018 International Workshop on Software Qualities and their Dependencies
 LTB 2018 Load Testing and Benchmarking of Software Systems
 ASBDA 2017 International Workshop on Autonomic Systems for Big Data Analytics
 AMS 2017 International Workshop on Architecting with MicroServices
 QUDOS 2017 International Workshop on Quality-Aware DevOps
 LTB 2017 Load Testing and Benchmarking of Software Systems
 QUORS 2017 International COMPSAC Workshop on Quality Oriented Reuse of Software
 LTB 2016 Load Testing and Benchmarking of Software Systems
 QUORS 2016 International COMPSAC Workshop on Quality Oriented Reuse of Software

JOURNAL REVIEWS

Brackets indicate the number of papers I reviewed (excluding revisions).

TSE (15) IEEE Transactions on Software Engineering
 TOSEM (15) ACM Transactions on Software Engineering and Methodology
 TAAS (10) ACM Transactions on Autonomous and Adaptive Systems
 Software (7) IEEE Software
 ROBOT (3) Elsevier Robotics and Autonomous Systems
 TSC (3) IEEE Transactions on Service Computing
 TCC (3) IEEE Transactions on Cloud Computing
 SPI (3) Wiley Software Process: Improvement and Practice
 IST (3) Elsevier Information and Software Technology
 Computing (3) Springer Computing
 JPDC (3) Journal of Parallel and Distributed Computing
 EMSE (3) Springer Empirical Software Engineering
 SoSyM (2) Springer Software & Systems Modeling
 Oxf Comp Jrnal (2) Oxford Academic Computer Journal
 JSEP (2) Wiley Journal of Software: Evolution and Process

SPE (2) Wiley Software: Practice and Experience
 Cloud (2) IEEE Cloud Computing
 IET Software (2) IET Software
 PLOS ONE (2) PLOS ONE
 ToMPECS (1) ACM Trans. on Modeling and Performance Evaluation of Computing Systems
 CSUR (1) ACM Computing Surveys
 Micro (1) IEEE Micro
 ASC (1) Elsevier Applied Soft Computing
 Computer (1) IEEE Computer
 Computing (1) IEEE Internet Computing
 JNCA (1) Elsevier Journal of Network and Computer Applications
 TNSM (1) IEEE Transactions on Network and Service Management
 JCST (1) Springer Journal of Computer Science and Technology
 JCC (1) Springer Journal of Cloud Computing
 SOCA (1) Springer Service Oriented Computing and Applications
 JSA (1) Elsevier Journal of Systems Architecture
 JBACS (1) Springer Journal of the Brazilian Computer Society
 JSS (1) Elsevier Journal of Systems and Software
 JWE (1) Journal of Web Engineering
 IBM (1) IBM Journal of Research and Development
 Computers (1) MDPI Computers
 Entropy (1) MDPI Entropy
 Supercomp. (1) Springer Journal of Supercomputing
 CACM (1) Communications of the ACM
 AIJ (1) Elsevier Artificial Intelligence Journal
 Neurocomp (1) Elsevier Neurocomputing
 Access (1) IEEE Access
 Neurocomp (1) Frontiers in Epidemiology
 TECS (1) ACM Transactions on Embedded Computing Systems

GRANT PROPOSAL REVIEW

2023 Swiss National Science Foundation (SNSF)
 2022 German Research Foundation, Deutsche Forschungsgemeinschaft (DFG)
 2021 National Science Foundation (NSF) panel (CISE/CNS)
 2021 National Science Foundation (NSF) panel (CISE/SHF)
 2020 National Science Foundation (NSF) panel (CISE/IIS)
 2019 Dutch Research Council (NWO)
 2018 Canadian Science Fund (FRQnet)
 2018 Austrian Science Fund (FWF)
 2016 Dutch Technology Foundation (STW)
 2023 Nazarbayev University Faculty-Development Competitive Research Grants

OTHER REVIEWS

2016 Elsevier book proposal review

INDUSTRY SERVICES

- SPEC I have been actively collaborating with the DevOps RG group to develop a DevOps framework by consolidating tools to better integrate performance monitoring and architectural refactoring.
- Intel and Microsoft During my postdoctoral research in IC4 (Irish cloud center with 40+ industry members), I was actively collaborating with Giovanni Estrada and Chris Woods from Intel and Niall Moran from Microsoft for developing auto-scaling mechanisms for OpenStack and Azure platforms, see [C15, C14, C18].
- IPMA Project Management Excellence My responsibility was to *assess* the quality of the projects submitted for the “National Project Management Excellence Award” and to judge their excellence by exploiting the IPMA Project Excellence Model. The assessment process included individual assessments, consensus meetings, site visits and report writing.
- Apache I am active in the Apache Storm community contributing to the auto-scaling feature of the framework: 🌐 <https://issues.apache.org/jira/browse/STORM-594>
- Imperial Consultants I was a research consultant on Big Data and Machine Learning in Imperial Consultants, a self-funding, wholly owned company of Imperial College London.

MEMBERSHIP IN TECHNICAL COMMUNITIES

- 2019– AAAI
- 2015– SPEC RG DevOps Performance Group
- 2011– IEEE, ACM, ACM SIGSOFT

IMMIGRATION STATUS

US Permanent Resident (Green Card holder since April 2020).

REFERENCES

- Christian Kästner** Associate Professor, Carnegie Mellon University, PA, USA.
Postdoc Advisor ☎ +1 412-268-5254 ✉ kaestner@cs.cmu.edu
🌐 <https://www.cs.cmu.edu/~ckaestne/>
- Claus Pahl** Professor, Free University of Bozen-Bolzano, Italy.
Ph.D. Advisor ☎ +39 0471-016-177 ✉ claus.pahl@unibz.it
🌐 <https://www.inf.unibz.it/~cpahl/>
- Baishakhi Ray** Associate Professor, Columbia University, NY, USA.
Collaborator ☎ +1 212-939-7112 ✉ rayb@cs.columbia.edu
🌐 <https://www.rayb.info/>
- David Garlan** Professor, Carnegie Mellon University, PA, USA.
Mentor & Collaborator ☎ +1 412-268-5056 ✉ garlan@cs.cmu.edu
🌐 <https://www.cs.cmu.edu/~garlan/>
- Jason O’Kane** Professor, Texas A&M, Texas, USA.
Mentor & Collaborator ☎ +39 0471-016-177 ✉ jokane@tamu.edu
🌐 <https://jokane.net/>
- Norbert Siegmund** Professor, Leipzig University, Germany.
Collaborator ☎ +49 3643-58-35-74 ✉ norbert.siegmund@uni-leipzig.de
🌐 <https://sws.informatik.uni-leipzig.de/>
- Sven Apel** Professor, Leipzig University, Germany.
Collaborator ☎ +49 3643-58-35-74 ✉ apel@cs.uni-saarland.de
🌐 <https://www.se.cs.uni-saarland.de/apel/>
- Ladan Tahvildari** Professor, University of Waterloo, Canada.
Mentor ☎ +1 519-888-4567 ✉ ladan.tahvildari@uwaterloo.ca
🌐 <https://uwaterloo.ca/electrical-computer-engineering/profile/ltahvild>
- Isil Dillig** Professor, UT Austin, USA.
Mentor ☎ +1 (512) 471-9794 ✉ isil@cs.utexas.edu
🌐 <https://www.cs.utexas.edu/~isil/>
- Marco Valtorta** Professor, University of South Carolina, SC, USA.
Mentor & Collaborator ☎ +1 803-777-4641 ✉ mgv@cse.sc.edu
🌐 <https://www.cse.sc.edu/~mgv/>